

CCSU BCA Semester 2 – Data Structures

2026 Exam-Oriented Important Topics + Short Notes

1. Introduction

Short Note: A data structure is a way of organizing and storing data so that operations can be performed efficiently. Data structures are broadly linear and non-linear.

Exam focus: Definition + key points + example/diagram/program where applicable.

2. Arrays

Short Note: An array stores same-type elements in contiguous memory locations. Access by index is fast. Learn traversal, insertion, deletion, searching and basic sorting.

Exam focus: Definition + key points + example/diagram/program where applicable.

3. Linked List

Short Note: A linked list consists of nodes containing data and a link to the next node. Types include singly, doubly and circular linked lists. Insertion/deletion can be easier than arrays when links are available.

Exam focus: Definition + key points + example/diagram/program where applicable.

4. Stack

Short Note: Stack follows LIFO (Last In, First Out). Main operations are push, pop and peek/top. Applications include recursion, expression evaluation and undo operations.

Exam focus: Definition + key points + example/diagram/program where applicable.

5. Queue

Short Note: Queue follows FIFO (First In, First Out). Main operations are enqueue and dequeue. Circular queue avoids wasted space; priority queue serves elements according to priority.

Exam focus: Definition + key points + example/diagram/program where applicable.

6. Trees

Short Note: A tree is a non-linear hierarchical structure. Learn terminology: root, parent, child, leaf, height and depth. Binary tree has at most two children per node.

Exam focus: Definition + key points + example/diagram/program where applicable.

7. Binary Search Tree

Short Note: A BST keeps smaller keys in the left subtree and larger keys in the right subtree. Average search/insert/delete can be efficient, but an unbalanced tree can become inefficient.

Exam focus: Definition + key points + example/diagram/program where applicable.

8. Graphs

Short Note: A graph consists of vertices and edges. Learn directed/undirected graphs, adjacency matrix/list, degree, path and traversal.

Exam focus: Definition + key points + example/diagram/program where applicable.

9. Searching & Sorting

Short Note: Linear search checks elements sequentially. Binary search repeatedly halves a sorted search space. Learn bubble, selection, insertion and merge sort basics, with time-complexity comparison.

Exam focus: Definition + key points + example/diagram/program where applicable.

■ Important Questions

- Explain **Introduction** with suitable example/diagram/formula/program.
- Explain **Arrays** with suitable example/diagram/formula/program.
- Explain **Linked List** with suitable example/diagram/formula/program.
- Explain **Stack** with suitable example/diagram/formula/program.
- Explain **Queue** with suitable example/diagram/formula/program.
- Explain **Trees** with suitable example/diagram/formula/program.
- Explain **Binary Search Tree** with suitable example/diagram/formula/program.
- Explain **Graphs** with suitable example/diagram/formula/program.
- Explain **Searching & Sorting** with suitable example/diagram/formula/program.

Note: These are revision notes aligned to the CCSU BCA curriculum structure available for 2025–26. Check your college's paper combination/electives before the exam.